230143V
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Lab – MDP

Exercise 01

State	Expected Utility (North)	Expected Utility (East)	Expected Utility (South)	Expected Utility (West)	Expected Utility (Nothing)	Best Action	Expected Utility for Best Action	Updated Utility
1	0	0	0	0	0	ANY	0	-0.1
2	0	0	0	0	0	ANY	0	-0.1
3	0	0	0	0	0	ANY	0	1
4	0	0	0	0	0	ANY	0	-0.1
5	0	0	0	0	0	ANY	0	-0.1
6	0	0	0	0	0	ANY	0	-0.05

1	-0.1	-0.1	-0.1	-0.1	-0.1	ANY	-0.1	-0.1999
2	-0.045	0.89	-0.045	-0.1	-0.1	EAST	0.89	0.7891
3	0	0.9475	0945	-0.0425	1	STAY	1	1
4	-0.1	-0.1	-0.1	-0.1	-0.1	NORTH	-0.1	-0.1999
5	-0.0975	-0.055	-0.0975	-0.1	-0.1	EAST	-0.055	-0.1549
6	-0.0525	0.0025	0.8925	-0.0425	-0.05	SOUTH	0.8925	0.8416

1	-0.1504	0.6902	-0.1504	-0.1999	-0.1999	EAST	0.6902	0.5895
2	-0.0994	0.9317	0.7502	-0.1482	0.7891	EAST	0.9317	0.8308
3	0.8469	0.9921	0.9895	0.8023	1	STAY	1	1
4	-0.1976	-0.1594	-0.1977	-0.1999	-0.1999	EAST	-0.1594	-0.2592
5	-0.1074	0.7892	0.7423	-0.1482	-0.1549	EAST	0.7892	0.6884
6	0.7918	0.8495	0.9343	-0.0474	0.8416	SOUTH	0.9343	0.8834

Exercise 02

[East, East, Stay, East, East, South]

Exercise 03

Output   Clear

```
[[['east'], ['east'], ['south']], [['east'], ['east'], ['stay']]]
graph at convergence:
[0.7191, 0.8283, 0.9373]
[0.7698, 0.8847, 1]

Best Policy:
[['east'], ['east'], ['south']]
[['east'], ['east'], ['stay']]

Policy converged at iteration: 3
graph converged after: 11 iterations

=== Code Execution Successful ===
```